

Since  $T = 1/f = 1/50 = 20 \text{ ms}$ ,  
5 ms is  $\frac{1}{4}$  of a period.

Correct diagram

[1]

- (b) Sound waves is directed from a loudspeaker towards a metal plate. A microphone, connected to a cathode ray oscilloscope (CRO), is placed in between the loudspeaker and the metal plate as shown in Fig. 4.3.

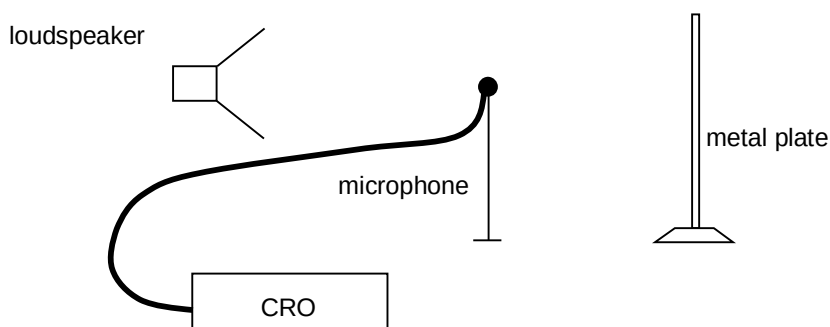


Fig. 4.3

- (i) Explain how stationary waves are formed in between the loudspeaker and the metal phase.

As the sound waves reflects off the metal plate, the incident and reflected waves have the same amplitude, wavelength and frequency [B1] move in opposite direction and overlap/superpose/meet [B1].

When they experience constructive interference, they form regions of high intensity and when they experience destructive interference, they forms regions of low intensity. [B1]

[3]

- (ii) The trace on the CRO in Fig. 4.4 shows the variation of signal with time at an antinode.

The time-base setting on the CRO is  $0.10 \text{ ms cm}^{-1}$ .

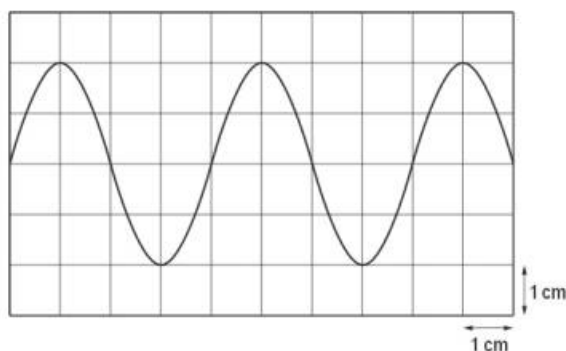


Fig. 4.4

The microphone is then moved by 3.3 cm and the trace on the CRO now records zero amplitude.

1. Determine the frequency of the sound.

$$\text{Frequency} = 1/\text{period} = 1 / 4 \times 10^{-3} = 2500 \text{ Hz} \quad [\text{A1}]$$

frequency = ..... Hz [1]

2. Calculate the speed of sound.

Distance between node and antinode =  $\frac{1}{4}$  wavelength.

$$\text{Wavelength} = 4 \times 3.3 = 13.2 \text{ m} \quad [\text{M1}]$$

$$v = f\lambda = 2500 \times 13.2 = 330 \text{ m s}^{-1} \quad [\text{A1}]$$

speed = ..... m s<sup>-1</sup> [2]

- 5 (a) For a particular gas, the emission and absorption spectra are obtained for the visible light spectrum.

Discuss one similarity and one difference between the discrete lines of the absorption and emission spectra of this gas.

..... {Similarity} The discrete lines of both absorption and emission spectrum .....  
 ..... occur at same frequencies [1], but {Difference} the absorption has dark .....  
 ..... lines against a continuous spectrum whereas the emission spectrum has .....  
 ..... coloured lines against a black background [1]. .....  
 ..... [2]

- (b) Fig. 5.1 gives information on three lines observed in the emission spectrum of hydrogen atoms.

| wavelength / nm | energy of photon / eV |
|-----------------|-----------------------|
| 486             | 2.56                  |
| 656             | 1.90                  |
| 1880            | 0.66                  |

Fig. 5.1

- (i) Complete Fig. 5.1 by calculating the energy of the photon with wavelength 656 nm. [1]
- (ii) Fig. 5.2 is a partially completed diagram to show energy levels of a hydrogen atom.

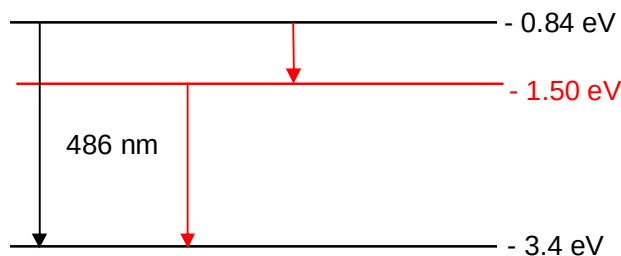
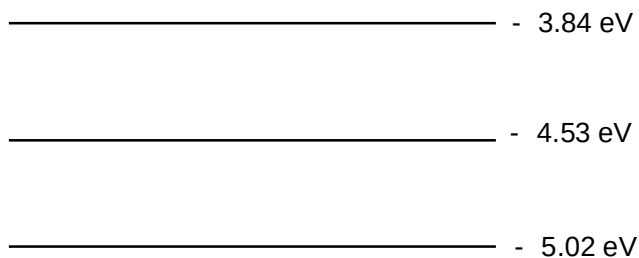


Fig. 5.2

On Fig. 5.2, draw an additional labelled energy level which will account for the emission of the photons in Fig. 5.1. [1]

- (c) Three of the energy levels of a lithium atom are shown in Fig. 5.3.



**Fig. 5.3**

One way to study the energy levels of an atom is to bombard the atom with electrons and measure the kinetic energies of the bombarding electrons before and after the collision. If a lithium atom which is originally in the  $-5.02$  eV level is bombarded with an electron of kinetic energy  $0.92$  eV, the scattered electron can have only two possible kinetic energies.

States these two kinetic energy values, and state what happens to the lithium atom in each case.

1<sup>st</sup> possible kinetic energy value: ..... eV [1]

0.92 eV [1] .....  
Nothing happens to the lithium atom (ie. it stays in the  $-5.02$  eV level) [1] ..... [1]

2<sup>nd</sup> possible kinetic energy value: ..... eV [1]

$0.92 - (5.02 - 4.53) = 0.43$  eV [1] .....  
Atom will be excited to the  $-4.53$  eV level [1] ..... [1]

6 (a) A stationary nucleus of a radioactive nuclide  $^{218}\text{Po}$  underwent a chain of decays by the